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Data Driven-Machine Learning-Based Fraud Detection Models in FinTech Financial Transactions

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***Abstract*—Fraud examination recognizes and combats improper economic activities by applying data analysis methods and machine learning for the purpose of improving safety and reducing financial threats. Financial fraud plays a big part in the FinTech sector and impacts many people and businesses all around the world. The development of automated fraud detection systems is ongoing, but reliably and rapidly identifying fraudulent transactions is still a significant challenge. The credit card fraud detection dataset from Kaggle is utilized in this study to train the XGBoost model. The dataset receives in-depth preprocessing treatment where researchers perform missing value imputation and handle outliers alongside class imbalance through the implementation of SMOTE. The XGBoost model is evaluated using a variety of significant performance metrics, including accuracy, precision, recall, and AUC. XGBoost delivers superior outcomes than traditional Decision Tree classifiers as well as SVM and Logistic Regression by reaching 99.32% accuracy alongside 99.31% precision and 99.42% recall and 99.7% AUC. The model demonstrates exceptional capability to detect fraudulent transactions because of its superior accuracy rates. The examined research creates stronger financial fraud detection methods for digital systems to protect secure transactions and minimize monetary damages from fraudulent operations.**

Keywords—Financial fraud detection, Fintech industry, Machine learning, XGBoost, Credit Card fraud detection dataset.

I. Introduction

Financial technology, or commonly known as FinTech, encompasses the use of technological applications in the financial service industry in a bid to increase the effectiveness and convenience of the services being offered. These range from e-commerce payments, digital credit, online lending, robo-advice, and blockchain-based transactions, among others, which have revolutionized the entire financial sector [1]. Mobile banking has been on the rise as the use of mobile phones to facilitate payment solutions made easier thus reducing the attachment to banking infrastructure [2]. Nevertheless, the increase of technological growth also brought the risk of financial fraud into the sphere of digital financial services. Individuals, companies, and banks throughout the world are all vulnerable to financial fraud. Fraudulent activities like identity theft, credit card fraud [3], money laundering, phishing scams, and insider trading have become more sophisticated, exploiting security loopholes in financial systems. Fraud attempts are becoming more sophisticated and commonplace as the volume of digital transactions increases, costing businesses a lot of money and hurting their reputations. As more and more individuals use digital payments and automated financial services, a real-time fraud detection and prevention system is essential [4].

These days, regulators, financial institutions, and FinTech startups all agree that detecting financial fraud is a primary priority [5][6]. Despite their limited effectiveness, conventional rule-based fraud detection methods cannot keep pace with the ever-evolving fraud tactics [7][8]. As financial fraud continues to evolve, leveraging advanced AI-powered detection mechanisms remains crucial for safeguarding digital transactions and maintaining trust in FinTech solutions [9]. As a result, data-driven approaches leveraging AI [10] and ML have gained prominence in fraud detection systems [11]. ML-based fraud detection algorithms sift through mountains of transaction data in search of irregularities, outliers, and unusual actions that could point to fraud. Deep learning (DL) further enhances detection capabilities by processing complex, high-dimensional data for real-time fraud identification.

A. Motivation and Contribution of the Study

The credit card fraud issue, which causes massive financial losses and damages consumer trust, is still very much alive and well in the financial sector. Unbalanced datasets, high-dimensional feature spaces, and changing fraud trends are common challenges for traditional fraud detection algorithms. Machine learning techniques, particularly advanced classification models, offer a promising solution by leveraging data-driven approaches to detect fraudulent transactions with high accuracy. This study aims to enhance fraud detection by employing XGBoost, a powerful ensemble learning model, to improve prediction performance while addressing key challenges such as class imbalance and feature selection. This study makes the following contributions:

- The suggested fraud detection method is guaranteed to be practically applicable as it makes use of the Kaggle Credit Card Fraud Detection dataset, which includes real transaction information.
- Establishes a reliable preparation pipeline to guarantee high-quality and consistent data by executing tasks such as normalization, outlier elimination using the Interquartile Range technique, and missing value imputation.
- Uses SMOTE to level the playing field in the dataset, which enhances model learning and reduces majority-class bias.
- Deals with the intricacies of fraud detection in extraordinarily unbalanced datasets and improves classification accuracy using the XGBoost algorithm.
- Evaluate fraud detection skills thoroughly by measuring key performance indicators like accuracy, precision, recall, and F1score.

B. Justification and Novelty

This work ensures practical applicability to financial fraud detection by using a real-world transactional dataset from Kaggle to support its methodology. The dataset is subjected to a thorough preparation pipeline that uses state-of-the-art

methods like SMOTE, imputation, and the Interquartile Range approach to tackle important issues including missing values, outliers, and extreme class imbalance. This study's originality comes from the fact that it applies XGBoost, a robust gradient boosting method, to the problem of fraud classification while simultaneously integrating feature normalization and selection techniques to promote model efficiency. Unlike traditional approaches, this research enhances fraud detection using data-driven insights, making it more scalable and resilient for use in real-world financial systems.

C. *Structure of the paper*

Here is the outline of the paper: Section II provides the background study on financial fraud detection, and Section III provides the research approach for this study. Section IV presents the experiment result and analysis of the model's performance. Section V conclusion and future work of study are provided.

II. Literature Review

This section of the report summarizes previous research on financial fraud detection and categorization. Classification methods were the primary emphasis of the literature studied. Some reviews are:

Wu et al. (2025) In India, fintech companies are increasingly gaining attention as the Government of India (GoI) is working to reach the underbanked people with additional financial services. The number of Indians having bank accounts has been on the increase, with estimates putting the proportion at close to 80%; these data useful for policymakers who are striving to include every person in the country's organized financial system [12].

Singh et al. (2024) machine learning techniques have proven effective in detecting fraudulent transactions. the performance of two well-known machine learning methods, such as XGBoost and ANNs, in identifying credit card fraud. Surpassing all five methods evaluated, XGBoost, with an accuracy of 92.7%, provides valuable insights for financial institutions looking to implement or improve fraud detection systems [13].

Aburbeian and Fernández-Veiga (2024) made a proposal for a system that would make online financial transactions safer by integrating ML with multi-factor authentication. The ML model was constructed by comparing 4 supervised classifiers: naive Bayes, LR, DT, and RF. According to the findings, the accuracy rate was 92.354% [14].

Chang et al. (2024) examine the effectiveness of two methods using SMOTE in detecting fraudulent transactions in unbalanced actual datasets: random under-sampling and over-sampling. The results show that random under-sampling has a strong re-call (92.86%) but poor precision, whereas SMOTE has a better recall (86.75%) and a more consistent F1score (73.47%) but slightly lower precision since at least two checks must be in place for a transaction to be considered really fraudulent [15].

Kumar P and Sountharrajan (2024) fraud detection technology coupled with ML algorithms utilized to assess the credibility of transactions. To enhance the detection accuracy, the system incorporates SVC and the MCC-based merchant profiles. The findings indicated that the use of MCC-based profiles in the category of Support Vector Machines (SVM) achieved an accuracy of 91 %, which is remarkably higher than basic methods [16].

Chung and Lee (2023) as online shopping and the use of IoT devices continue to explode in popularity, the issue of fraudulent credit card transactions has come to the forefront of people's minds throughout the world. Three ML models—linear regression, KNN, and linear discriminant analysis are used in their method. With this recommended method, the features extracted had an accuracy of 0.9362 [17].

The results, limitations, and potential directions for further research are compared in Table I.

TABLE I. Comparative Analysis of Machine Learning Approaches for Financial Fraud Detection

Author	Dataset	Methodology	Findings	Limitations	Future Work
Singh et al. (2024)	Publicly available credit card transaction dataset	XGBoost, Artificial Neural Networks (ANNs), Confusion Matrix, Performance Metrics	XGBoost achieved 92.7% accuracy, outperforming other models for fraud detection	Limited to two ML methods, lacks real-time testing	Extend to deep learning models and real-time fraud detection
Aburbeian and Fernández-Veiga (2024)	Digital financial transactions dataset	Multi-factor authentication + ML (Logistic Regression, Decision Trees, RF, Naïve Bayes)	Accuracy of 92.35% with two-layer security, including facial recognition	Dependence on facial recognition, limited classifier set	Test with other ML models and evaluate real-world deployment
Chang et al. (2024)	Imbalanced real-world financial dataset	Random Under-Sampling, SMOTE, Performance Metrics	SMOTE improved F1 score (73.47%), random under-sampling had high recall (92.86%)	Trade-off between precision and recall, limited to two resampling techniques	Test hybrid sampling techniques and real-time fraud detection
Kumar P and Sountharrajan (2024)	Synthetic dataset (50,000 transactions)	Support Vector Classifier (SVC), MCC-based merchant profiling, Decision Trees, RF	SVC achieved 91% accuracy, outperforming basic models	Synthetic dataset lacks real-world complexities	Apply MCC profiling to real bank data and improve model robustness
Chung and Lee (2023)	Credit card transaction dataset	KNN, Linear Discriminant Analysis, Linear Regression + Rule-based enhancements	Achieved 93.62% accuracy with conditional statements for better fraud detection	Lack of scalability for large datasets	Implement deep learning-based fraud detection for scalability
Wu et al. (2025)	RBI secondary data (India)	Regression, Correlation Analysis	FinTech expansion significantly increased financial inclusion in India	Limited to secondary data, no real-time financial fraud detection analysis	Expand to digital lending and mobile banking fraud detection

III. METHODOLOGY

Data obtained from the Kaggle Credit Card Fraud Detection dataset, which includes transactions from European credit cards, is the first step in the structured approach that comprises the research methodology for this study. For better model performance, the dataset is heavily preprocessed to handle missing values (imputation), outliers (Interquartile Range method), and class imbalance (SMOTE). Numeric features are normalized to [0,1] for consistency, and feature selection enhances efficiency and accuracy. The dataset is split into 70% training and 30% testing for model evaluation. For the classification, apply the XGBoost model to detect fraudulent transactions. Evaluation criteria like F1score, recall, accuracy, and precision are utilized to measure a performance of the model. The following steps are shown in Figure 1.

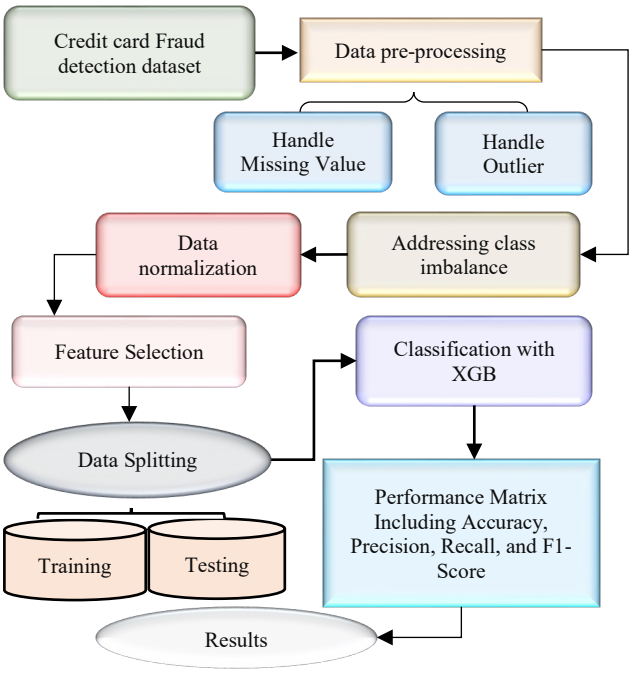

Fig. 1. Flowchart for Financial Fraud Detection

The following is a flow chart showing the main procedures for detecting financial fraud:

A. Data Collection

This investigation into credit card theft utilizes data collected by Kaggle Credit Card Fraud Detection. It included September 2013 transactions from credit cards throughout Europe. The data shows that 492 (0.172%) of the 284,807 transactions are fraudulent, making the positive class noticeably under-represented. Figure 2 represents the class distribution of data.

Fig. 2. Bar Chart for Class Distribution

Figure 2 shows a schematic depicting the data sample. Furthermore, there is a set of 'Class' tags included in every transaction. The usual set of tags is 0, while the fraudulent set is 1. A mere 492 transactions, or 0.172 percent of the whole dataset, fall into this category. A comparison of the dataset's positive and negative samples reveals a statistically significant difference (Figure 3).

Figure 3 presents the KDE curve of the dataset, illustrating the distribution of fraud and normal transaction scores. The red curve represents normal transactions, showing a higher density in the lower score range, whereas the blue curve corresponds to fraudulent transactions, which exhibit a peak at a relatively higher score range. The overlapping regions highlight potential classification challenges, emphasizing the necessity of robust fraud detection models to effectively differentiate between legitimate and fraudulent activities.

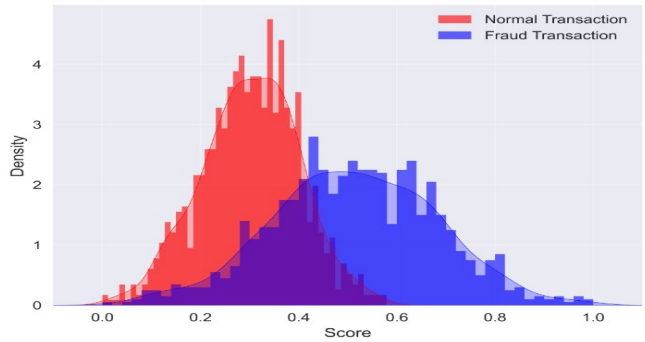

Fig. 3. Kernel Density curve of Dataset

B. Data Preprocessing

Data preprocessing ensures data quality by handling missing values, duplicates, class imbalance, and normalization. Initially, missing values, and data types, followed by appropriate imputation or removal of missing data. Outlier are eliminated to maintain integrity, and class imbalance is addressed to balance distributions. Numeric features are data normalized, and feature selection is performed to retain the most relevant attributes, improving model efficiency. The following procedures are outlined for the pre-processing phase:

- **Handle missing value:** missing data in the context, appropriate strategies such as imputation (using mean, median, mode, or predictive modeling) or removing rows/columns with significant missing values are chosen.
- **Handle outlier:** Any really out-of-the-ordinary results were handled using the Interquartile Range Method. It improved the outlier treatment procedure by making threshold modifications.

C. Addressing Class Imbalance

When the dataset is tiny, the Smote approach works effectively. Creating synthetic data points will be more time-consuming and inefficient when dealing with datasets that are big enough [18]. The Smote technique is used to construct a new synthetic data point, A, which is specified as Equation (1):

$$A = X + c(X_R - X) \tag{1}$$

Where X is a piece of data in the dataset, X_R is the one data point from the KNN of X that was chosen at random, and c (0 ≤ c ≤ 1) is a constant. The balancing pie chart is shown in Figure 4.

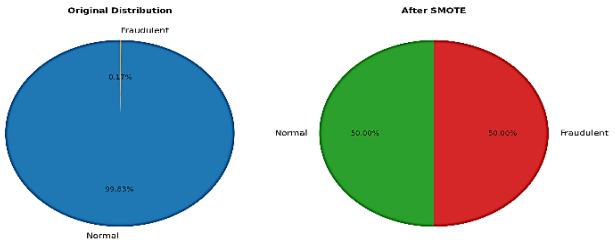

Fig. 4. Pie Chart for Data Balancing

Figure 4 shows the class distribution before and after applying SMOTE. Normally, transactions dominate (99.83%), while fraudulent ones are only 0.17%, causing class imbalance. After SMOTE, both classes are balanced (50% each), enhancing fraud detection and model performance.

D. Data Normalization

The clustering and classification models are made more efficient by normalizing the numerical data such that all values fall within a defined range [19]. It changes the range of each numerical attribute to [0, 1]. The transformation is achieved through Equation (2):

$$X' = \frac{(X - Xmin)}{(Xmax - Xmin)} \tag{2}$$

where X signifies the original value, X' represents the normalized value, and $Xmax$ and $Xmin$ denote the maximum and minimum values of the respective feature.

E. Feature Selection

In computer vision and ML applications, feature extraction strategies and tactics are crucial because it allow the conversion of unstructured data into meaningful representations [20]. PCA and other dimensionality reduction approaches preserve the most useful features while reducing complexity and improving discriminative capability.

F. Data Splitting

The dataset was divided into two parts during preprocessing: the training set is utilized to train the model with 70% of the data, and the testing set is utilized to assess the model's performance with 30% of the data.

G. Classification with XGBoost Model

The exceptional performance of XGBoost algorithms has led to their recent rise in popularity as a state-of-the-art tool for classification and regression problems. In 2016, Chen and Guestrin developed a method for scalable tree boosting called the XGBoost algorithm. It is crucial to note that the approach takes into account the dataset's missing values, the dataset's abundance of zeroes, and the results of the feature engineering. Because XGBoost only employs DT as its base classifiers, the complexity of the trees is controlled by a modified loss function in Equation (3,4).

$$L_{xgb} = \sum_{i=1}^{N} L\big(y_i, F(x_i)\big) + \sum_{m=1}^{M} \Omega(h_m) \tag{3}$$

$$\Omega(h) = \gamma T + \frac{1}{2}\lambda \parallel \mathcal{W} \parallel^2 \tag{4}$$

The total number of tree leaves is represented by T, and their corresponding output ratings are denoted by w. This loss function may be used as part of the decision tree's split criteria for a pre-pruning strategy. The output of less complicated trees is caused by raising the value of γ. The minimum gain in loss reduction needed to split an internal node is determined by the γ value. Applying XGBoost's shrinkage regularization parameter decreases the additive expansion's step size. Managing the tree depth is another tactic that might reduce complexity. Additional advantages of reducing tree complexity include faster model training and reduced storage space needs.

H. Performance Metrics

Evaluation metrics are chosen according to how well it address the issue at hand. Measures like as recall, precision, F-measure [16] and ROC area are often used to evaluate classifier performance in CCFD situations. These metrics are utilized to quantify how well each classifier performs.

Accuracy: The proportion of properly categorized occurrences to all classified instances is known as accuracy. It is expressed Equation (5):

$$Accuracy = \frac{\text{TN} + \text{TP}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{5}$$

- **True positive (TP):** the total amount of cases where fraudulent transactions were identified.
- **True negative (TN):** the quantity of legal transactions that are deemed to be legal
- **False positive (FP):** the amount of valid transactions that were stigmatized as fraudulent.
- **False negative (FN):** the observation is really benign, but models mistakenly anticipate it as malignant due to the high percentage of fraudulent transactions that are labeled as legal.

Precision: The precision is a measure of the percentage of correctly classified positive cases as a percentage of all positive occurrences. It is represented as Equation (6):

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \tag{6}$$

Recall: The ratio of categorized positive cases to total positive instances is known as recall. Best recall is indicated if the ratio's value is 1, as opposed to 0. It is represented Equation (7):

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{7}$$

F1-score: The F1-score is a synthetic metric for recall and precision [21]. This metric performs better than accuracy when dealing with imbalanced datasets. The F1 score can be defined as follows in Equation (8):

$$\text{F1} = \frac{2*(\text{precision}*\text{recall})}{\text{precision+recall}} \tag{8}$$

ROC-curve: To assess the model's class discrimination capabilities, it plot the ROC curve and AUC score.

IV. Results & Discussion

The experimental equipment utilized for CCFD is a CPU variation of 1050 H, 8 GB of RAM, 240 GB of SSD, and the Python v3 language. The goal is to train and evaluate the model in the shortest amount of time possible. Several categorization techniques were employed in this investigation, and their results are examined in this section. Table II shows the results of the evaluations done on the XGB model using a dataset for CCFD. Metrics such as F1score, recall, accuracy, and precision were used to evaluate the models.

TABLE II. XGB MODEL PERFORMANCE ON CREDIT CARD FRAUD DETECTION DATASET

Evaluation Matrix	Extreme Gradient Boosting (XGB)
Accuracy	99.32
precision	99.31
Recall	99.42
F1-score	99.31
ROC-AUC	99.7

A performance of a proposed XGB model is presented through bar graphs in Figure 5 and in Table II. The performance measurements of XGB are 99.32% of accuracy, 99.31 % of precision, and 99.42% of recall. The algorithmic's high accuracy, manifesting in the low levels of FP and FN instances, can be supported by the high values of 99.7 percent of ROC-AUC and 99.31 percent of F1-score. These findings have shown that the XGB is effective and consistent in classification duties.

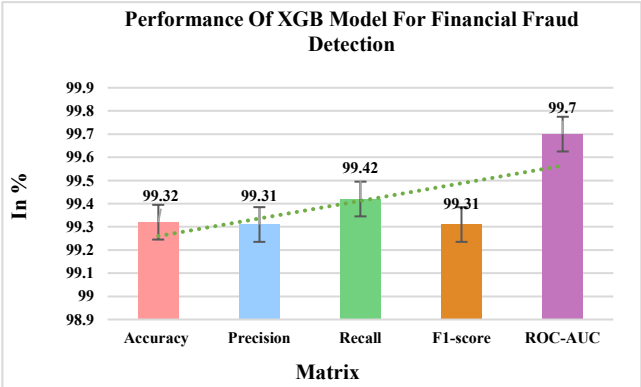

Fig. 5. Bar Graph for XGB Model Performance

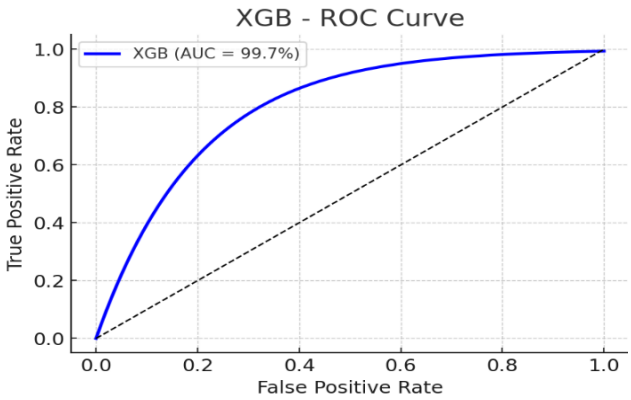

Fig. 6. ROC graph of XGBoost model

AUC of 99.7% clearly shows that the model has a nearly perfect classification ability, which is seen in Figure 6. The blue line indicates the performance of the model, while the black dotted line represents simply the random chance of classification. The model has a high value of AUC which indicates that the model has low FPR and high TPR making it suitable to be used in FinTech transactions for fraud detection.

	precision	recall	f1-score	support
0	0.990	0.997	0.993	7422
1	0.997	0.990	0.993	7406
accuracy			0.993	14828
macro avg	0.993	0.993	0.993	14828
weighted avg	0.993	0.993	0.993	14828

Fig. 7. Classification report of XGBoost model

Figure 7 displayed the classification report of the XGBoost model, presenting precision, recall, F1score, and support for both non-fraudulent (0) and fraudulent (1) transactions. For class 0, the model achieves a precision of 0.990, recall of 0.997, and F1 score of 0.993, with support of 7,422 instances. Similarly, for class 1, the precision is 0.997, recall is 0.990, and the F1score is 0.993, with a support of 7,406 instances. The detection model achieves 99.3% accuracy while maintaining effective detection of fraudulent along with non-fraudulent transactions with low misclassification rates.

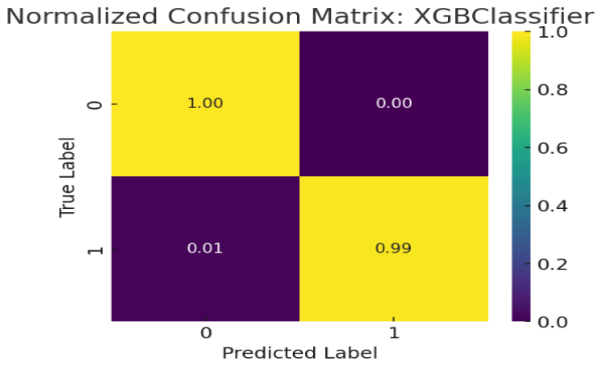

Fig. 8. Confusion Matrix of XGBoost Model

Figure 8 illustrates the XGBoost model's confusion matrix, which indicates how well it detects fraud. The classifier achieves 100% accuracy in identifying non-fraudulent transactions (TN) and correctly classifies 99% of fraudulent cases (TP). The FPR is 0%, indicating no legitimate transactions were misclassified as fraud, while the FNR is only 1%, meaning very few fraud cases were missed. The XGBoost model's ability to identify bogus financial transactions.

A. Comparative Analysis

The comparison between the proposed XGBoost and base model, including DT [22], SVM [23], and LR [12], performance is provided in this section. The following comparison is shown in Table III.

TABLE III. COMPARATIVE ANALYSIS FOR FINANCIAL FRAUD DETECTION BETWEEN BASE AND PROPOSE MODEL

Matrix	Accuracy	Precision	Recall	F1-score
DT[22]	88.16	95	81	87
SVM[23]	94.9	95.9	95.1	95.1
LR[12]	94.65	97.32	91.85	94.51
XGB	99.32	99.31	99.42	99.31

Table III presents a comparative analysis of financial fraud detection performance between baseline models and the proposed XGBoost (XGB) model. Decision Tree (DT) achieved an accuracy of 88.16%, with a precision of 95%, recall of 81%, and an F1score of 87. SVM performed better, attaining 94.9% accuracy, 95.9% precision, 95.1% recall, and an F1score of 95.1. LR showed similar results with 94.65% accuracy, 97.32% precision, 91.85% recall, and an F1score of 94.51. The proposed XGBoost model significantly outperformed all baselines, achieving the highest accuracy of 99.32%, along with 99.31% precision, 99.42% recall, and a 99.31 F1score, demonstrating its superior effectiveness in detecting financial fraud.

XGBoost-based fraud detection delivers various substantial benefits such as maximum accuracy together with imbalanced data resilience and rapid processing of big datasets. Through its ensemble learning technique, the

predictive model improves its performance by reducing both bias and variance. XGBoost exceeds conventional models because it generates better classification metrics which reduces FP and FN counts. The reliable properties of XGBoost make it an effective tool for financial fraud detection and increase digital transaction security.

V. Conclusion and Future Work

The global rise of financial fraud occurs predominantly in FinTech along with strong growth in digital transaction activities. Efficient fraud detection methods need development since fraudulent activity poses considerable threats to consumers along with financial institutions. A test of the XGBoost model occurs with the Kaggle CCFD dataset to evaluate its capability in detecting fraudulent transactions. The proposed model demonstrates superior performance because it reaches 99.32% accuracy alongside 99.31% precision and 99.42% recall while maintaining 99.7% AUC, better than Decision Tree, SVM and Logistic Regression baseline models. Yet despite these excellent performance results, the model faces limitations from its complex structure that could lead to overfitting and its usage of SMOTE for balancing classes, which may distort true fraud patterns. Further, the fact that the used dataset contains historical fraud patterns may not generalize well to the current and future types of fraud. As to future work, deep learning-based ensemble methods and real-time detection models will be investigated to improve the model's flexibility. On the same note, utilizing the explainable AI approach will help decide on the integration of the techniques to enhance the model explainability in financial security applications.

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